

Gayathri Marichetty Sudhakar

Senior Data Engineer | Cloud Data Platforms | Azure, AWS, Snowflake, Spark, Python

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Executive Summary

Senior Data Engineer with **5+ years** of experience building **cloud data platforms, data lakes, and data warehouses** across **Azure, AWS, Snowflake, and Databricks**. Expert in **batch** and **streaming pipelines** using **Spark, Kafka, Python, and SQL**, delivering analytics-ready data for **healthcare, energy, and enterprise analytics**. Proven record of improving **data reliability, performance, and cost optimization** while supporting **BI, ML, and regulatory reporting**. Actively leverage **AI-powered development and analytics tools** to accelerate **pipeline development**, improve **data quality**, and enhance **engineering productivity**.

- Designed and scaled **batch** and **event-driven data pipelines** using **Azure Data Factory, Databricks, Apache Spark, Kafka, Snowflake, BigQuery, and Azure Synapse**, delivering reliable, analytics-ready data across large enterprise environments.
- **Built and maintained Lakehouse** and **data warehouse layers** supporting **clinical, pharmacy, finance, and enterprise reporting**, creating a trusted source of truth for **leadership, compliance, and business teams**.
- Engineered **high-throughput ingestion pipelines** processing **millions of records** with **sub-minute latency**, powering **live dashboards, operational monitoring, and downstream AI/ML workloads**.
- Enhanced and optimized **data processing and orchestration frameworks** using **Python, Java, SQL, and ETL/ELT** patterns, enabling scalable transformations, validation, and automation.
- Architected **cloud data platforms** across **Azure, AWS, Snowflake, BigQuery, and Azure Synapse**, designing cost-efficient, scalable, and highly available data lakes and warehouses.
- Enabled **GenAI and AI-driven analytics** by preparing **feature-ready, governed datasets** and **pipelines** for **LLM and ML applications**, accelerating predictive insights.
- **Orchestrated deployments and CI/CD pipelines** using **Terraform, Docker, Kubernetes, and Jenkins**, ensuring secure, repeatable releases while **optimizing Spark workloads** and **cloud compute for 30–60% performance and cost gains**.
- Enforced **HIPAA-aligned data governance, quality checks, lineage, RBAC, and audit controls**, securing sensitive healthcare and enterprise data while maintaining regulatory compliance.
- Collaborated in **Agile teams** with analysts, clinicians, product managers, and engineering leaders to deliver **dashboards, insights, and AI/ML-ready data products**, improving business decision-making and analytics readiness.

Technical Skills

Cloud & Data Platforms: Azure, AWS, Snowflake, Databricks, GCP

Cloud Services: Azure Databricks, Azure Synapse, ADLS, AWS S3, AWS Lambda, AWS Glue, Google BigQuery, GCS

Data Engineering: ETL / ELT, Data Lake, Data Warehouse, Star Schema, Dimensional Modeling, SCD Type 1/2, Data Governance, Data Lineage.

Processing & Streaming: Apache Spark, PySpark, Apache Kafka

Pipelines & Orchestration: Azure Data Factory (ADF), Apache Airflow, dbt (ELT)

Analytics & Reporting: Tableau, Power BI, Looker, ggplot2, Business Intelligence

Languages: Python, SQL

Gen AI & AI Enablement: LLM- Ready Data Pipelines, Azure Open AI Integration, Vector Embeddings

DevOps & Infrastructure: CI/CD, Terraform, Docker, Kubernetes

Security & Governance: IAM, RBAC, Secrets Management, Data Quality, Data Lineage

Education

Master of Science in Information Technology - Florida International University, Miami, FL

Bachelor of Technology in Information Technology - Anna University, Chennai, India

Work Experience

KARE Health Partners – Senior Data Engineer (Jan 2025 – Present, Tampa, FL)

Healthcare technology company delivering data platforms and analytics for pharmacy and provider networks.

- Built **batch and real-time data pipelines** using **Azure Data Factory, Databricks, Spark, Kafka, Snowflake, BigQuery, and Azure Synapse**, delivering **analytics-ready data sets** across 5 enterprise domains, **improving data availability by 40%**.
- Established **low-latency, high-throughput ingestion frameworks** processing 10M daily records, accelerating **real-time insights for dashboards, operations, and AI/ML applications**.
- Developed **Spark and Scala** pipelines on **Hadoop (HDFS, Hive, MapReduce)** to transform healthcare data, improving data readiness by 50%.
- Automated **ETL workflows** using **Python, Java, Bash, and SQL**, reducing manual processing by 75%.
- Enabled **GenAI and AI-driven analytics initiatives** by preparing feature-ready, governed datasets for **downstream ML and LLM applications**, accelerating predictive analytics by **25%**.
- Implemented **CI/CD and Infrastructure-as-Code** using **Git, Jenkins, Docker, Kubernetes, and Terraform**, accelerated production releases by 50%, while ensuring secure and repeatable releases.
- Enforced **HIPAA-compliant data quality, governance, lineage, RBAC, and Audit controls**, reducing data errors by 35% and securing sensitive healthcare data.

NextEra Energy – Azure Data Engineer (Nov 2023- Dec 2024, Remote)

Clean energy leader using data and analytics to optimize power generation and grid operations.

- Built and optimized **Azure Data Lake, Snowflake, BigQuery, and Domo** pipelines integrating **CRM, ERP, and operational systems**, supporting more than 10 TB of enterprise data and improving availability by 40%.
- Designed and automated **ETL pipelines** using **Azure Data Factory, Informatica, Alteryx, and Python**.
- Developed high-performance SQL models across **Snowflake, Redshift, BigQuery, and Azure SQL**.
- Delivered **Power BI, Tableau, and Looker** dashboards with KPIs, increasing reporting efficiency by 50%.
- Configured **Spark and Hadoop** pipelines for data processing across structured and unstructured datasets.
- Applied **Python-based ML pipelines (Scikit-learn, TensorFlow, PySpark MLlib)** to generate predictive insights, improving model accuracy by up to 25%.

Abbott Laboratories –Data Engineer (Sep 2022 – Oct 2023, Orlando, FL)

Global healthcare company delivering diagnostics, medical devices, and data-driven clinical platforms.

- Built **Python and Snowflake ETL pipelines** processing 5 TB of **healthcare data** for analytics and reporting.
- Migrated **AWS workloads to serverless (Lambda, Kinesis)** using **Terraform and CloudFormation**.
- Developed **Spark, Hive, and MapReduce** pipelines to transform and load large datasets into **HDFS and HBase**.
- Enhanced **Kafka partitions** and replication, increasing streaming throughput by 30%.
- Implemented **Snowflake data models and SQL transformations** to support **clinical and operational reporting**.
- Integrated **API and third-party data sources** into centralized data pipelines, improving **data availability**.
- Built **CI/CD Pipelines and Dockerized environments**, reducing deployment cycles by 60%.

Florida International University – Graduate Research Assistant (May 2020 – Oct 2020, Miami, FL)

Public university using data science and analytics to study public health and COVID-19 trends.

- Analyzed large-scale **COVID-19 datasets** using **Python, SQL, and Pandas** to support public health research.
- Created **Tableau and Python** dashboards to track infection trends, vaccinations, and policy impact.
- Configured **data refresh and reporting**, cutting research turnaround time by 50%.

Nike – Data Engineer (June 2018 – Nov 2019, Bangalore, India)

Global sports and apparel company using cloud data platforms for retail and supply-chain analytics.

- Built **Redshift, Spark, and Hive ETL pipelines** for sales and distribution analytics.
- Optimized **AWS EMR and Lambda** workloads, reducing compute costs by 40%.
- Implemented **data quality and governance** controls across reporting pipelines.